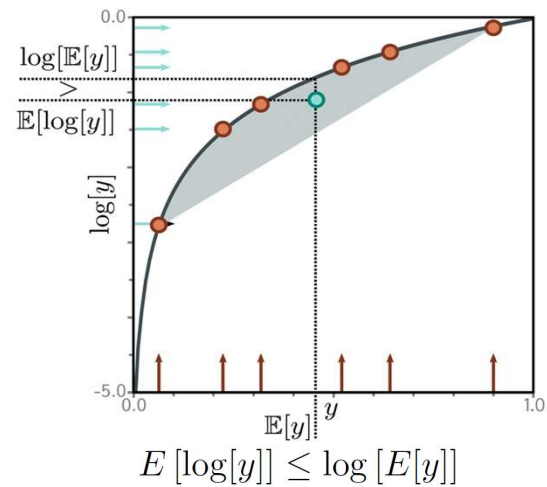


Agenda

- Last time
 - Inequalities (Markov, Chebyshev, Jensen)
- Today

Illustration for log function



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Markov's Inequality Discrete Case

$$\Pr[X > a] \leq \frac{E[X]}{a}$$

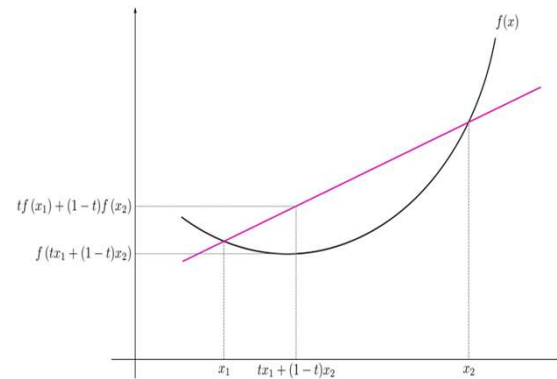
$$\begin{aligned}
 & a \Pr(X > a) \\
 = & a \Pr(X = a + 1) + a \Pr(X = a + 2) + a \Pr(X = a + 3) + \dots \\
 \leq & a \Pr(X = a) + (a + 1) \Pr(X = a + 1) + (a + 2) \Pr(X = a + 2) + (a + 3) \Pr(X = a + 3) + \dots \\
 \leq & \Pr(X = 1) + 2 \Pr(X = 2) + 3 \Pr(X = 3) + \dots + \\
 & a \Pr(X = a) + (a + 1) \Pr(X = a + 1) + (a + 2) \Pr(X = a + 2) + (a + 3) \Pr(X = a + 3) + \dots \\
 = & E(X)
 \end{aligned}$$

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Jensen's inequality

- For any concave function and random variable Y
 - $f[E[Y]] \geq E[f(Y)]$
- For a convex function
 - it's the other way around



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Example

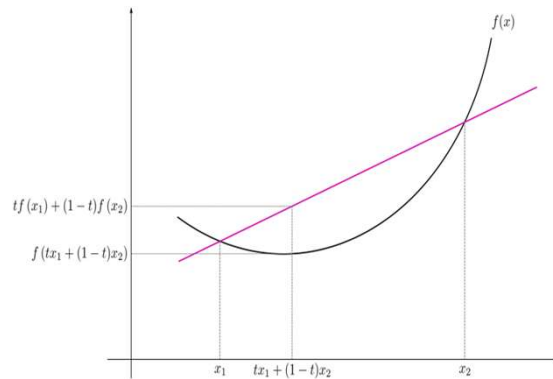


- $R = \begin{cases} 1.1 & \text{with probability } \frac{1}{2} \\ 0.9 & \text{with probability } \frac{1}{2} \end{cases}$ Interpret R as return on investment
 - $E[R] = 1$ suggests that there is no gain or loss in investment
- However, net return $N = R_1 R_2 R_3$
 - To find expected value of wealth, without assumption of independence, use
 - $\log N = \log R_1 + \log R_2 + \dots$
 - $E[\log N] = E[\log R_1] + E[\log R_2] + \dots$
 - From Jensen's formula on each term, $\text{LHS} < \log E[R_1] + \log E[R_2] + \dots$
 - If each return was, on an average 1, then $\text{LHS} < n \log E[R] = n \log 1 = 0$
 - Turning to the actual wealth $\exp\{E[\log N]\} < e^0 = 1$ return factor is negative
- Conclusion: It's a losing proposition, not a no-op proposition

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Agenda

- Last time
 - Inequalities
- Today
 - Weak Law of “large numbers”
 - Popular distribution functions



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Expected value

- When we say the expected value of a die roll is 3.5, what does that mean?
 - If you throw the die n times and find the sample mean, you should get a value close to 3.5 — provided n is large.
 - It says nothing about small n .
 - As n increases, the sample mean should move closer and closer to 3.5.
- That is the intuition. Let's state it more formally

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The weak law of large numbers

- Let $X_1, X_2, \dots, X_n$ be independent and identically distributed random variables, each with mean μ . For any $\varepsilon > 0$:

$$P\left\{\left|\frac{X_1 + X_2 + \dots + X_n}{n} - \mu\right| > \varepsilon\right\} \rightarrow 0 \text{ as } n \rightarrow \infty$$

- The **sample mean** converges in probability to the true mean

Must know

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The weak law

$$P\{|X - \mu| \geq k\} \leq \frac{\sigma^2}{k^2}$$

- From Chebyshev's inequality applied to the sample mean

$$E\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) = \mu, \quad \text{Var}\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

$$\therefore P\left\{\left|\frac{X_1 + X_2 + \dots + X_n}{n} - \mu\right| > \varepsilon\right\} \leq \frac{\sigma^2}{n\varepsilon^2}$$

- Letting $n \rightarrow \infty$ the right-hand side vanishes, so

$$\therefore \lim_{n \rightarrow \infty} P\left\{\left|\frac{X_1 + X_2 + \dots + X_n}{n} - \mu\right| > \varepsilon\right\} = 0$$

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Comments on the weak law

Nice to know

- The identically distributed assumption is not strictly necessary
 - The random variables just need the same mean μ .
 - We assume a common variance earlier
 - If not, the $n\sigma^2$ term becomes n times the expected variance
 - The proof still goes through.
- Independence is strictly not necessary
 - Pairwise independence suffice
- The law *does* assume a well-defined expected value and variance
 - Otherwise, it may not hold.

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Law of averages

- We tend to believe that if something has been going wrong for a while, it will suddenly turn right — by the “law of averages”
 - This supposed law is a fallacy
 - A gambler tosses an unbiased coin 20 times and gets a head every time
 - He invokes the “law of averages” and concludes the next toss is more likely to be a tail
 - The probability of getting 21 heads in a row is $(1/2)^{21}$
 - The probability of getting 20 heads followed by 1 tail is also $(1/2)^{21}$
 - The coin has no memory: Events are independent with identical distribution
 - However, if he keeps tossing the coin eventually he will see tails appearing, but the tails may not “compensate” for the first 20 heads
 - Relative frequency stabilize, past imbalance need not be cancelled

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